

04-14-2025

Panel Data:

→ Hansen 17-18; ch. from handbook

$$Y(0) \perp\!\!\!\perp D \mid (X, W)$$

$$\hookrightarrow Y(0) \perp\!\!\!\perp D \mid W$$

Linear Model for Untreated Potential Outcomes:

$$Y_i(0) = w_i^T \beta_0 + e_i$$

$$\text{Lin} \pm \text{unc} = E[e \mid W, D] = 0$$

$$ATT = E[Y \mid D=1] - E[w^T \mid D=1] \beta_0$$

→ if w known, linear adjustment

→ infeasible to do for $\hat{ATT}$ bc unknown

aside about omitted variable bias:

$$E[Y \mid D=1] - E[Y \mid D=0]?$$

$$= E[Y(1) - Y(0) \mid D=1] + E[Y(0) \mid D=1] - E[Y(0) \mid D=0]$$

→ add and subtract

$$= ATT + \underbrace{E[Y(0) \mid D=1] - E[Y(0) \mid D=0]}_{\text{selection bias}}$$

→ intrinsic differences left uncontrolled

$$= ATT + E[w^T \mid D=1] \beta_0 - E[w^T \mid D=0] \beta_0$$

$$= ATT + \underbrace{(E[w^T \mid D=1] - E[w^T \mid D=0]) \beta_0}_{\text{OVB}}$$

2 conditions where $OVB = 0$

1) mean w b/w groups is same

2) no effect on outcome

Panel Data w/ $T=2$:

$\{Y_{it}, Y_{iz}, D_{it}, D_{iz}\}_{i=1}^n$ iid across units

$\Delta Y_{it} := Y_{it} - Y_{it-1}$ (first differencing)

$\underline{Y}_i = \begin{pmatrix} Y_{i1} \\ Y_{i2} \end{pmatrix}$ (vector of y 's through time)

$\underline{D}_i = \begin{pmatrix} D_{i1} \\ D_{i2} \end{pmatrix}$ (same for treatment)

$\underline{d} = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$ eg $\underline{d} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ untreated in $t=1$, treated $t=2$
↳ treatment regime

Potential Outcomes

$Y_{it}(\underline{d})$ e.g. $Y_{iz}(0,0)$ untreated $\forall t \in T$

$\underline{Y}_i(\underline{d}) = \begin{pmatrix} Y_{i1}(\underline{d}) \\ Y_{i2}(\underline{d}) \end{pmatrix}$

Panel Data Natural Experiment:

- no units treated in $t=1$ s.t. $D_{i1}=0 \forall i$

⇒ in $t=2$ some units become treated

s.t. $D_{i2} = \begin{cases} 1 & \text{treat} \\ 0 & \text{untreat} \end{cases}$

- simplified notation:

G_i : group of a unit (time period when treated)

↳ $G_i = 2$ if $D_{i2}=1$

↳ $G_i = \infty$ if $D_{i2}=0$ (inf bc never true)

$Y_{it}(g), Y_{it}(\infty), \underline{Y}_i(g), \underline{Y}_i(\infty)$

→ untreated potential outcomes for $G=g$ or $G=\infty$

Observed outcomes

$\underline{Y}_i = \underline{Y}_i(G_i)$

No Anticipation Condition:

in pre-treatment periods s.t. $t < G_i$, $Y_{it} = Y_{it}(\infty)$

$$\text{ex: } G_i = 2 \Rightarrow Y_{i1} = Y_{i1}(2) = Y_{i1}(\infty)$$

Unconfoundedness:

$$- Y_i(\infty) \perp\!\!\!\perp G_i \mid \underline{W}_i$$

s.t.

$$\underline{W}_i = \begin{pmatrix} W_{i1} \\ W_{i2} \end{pmatrix}$$

Linear Model for Untreated Potential Outcomes

$$\begin{aligned} - Y_{it}(\infty) &= \underline{W}_{it}^T \beta_t + e_{it} \\ &= \theta_t + \underline{W}_{it}^T \beta_t + e_{it} \\ &\quad \hookrightarrow \text{intercept} \end{aligned}$$

$$\text{Lin + unc: } \Rightarrow E[e_t \mid \underline{W}, G] = 0$$

Suppose:

$$Y_{it}(\infty) = \theta_t + \underline{W}_i^T \beta + e_{it}$$

$\hookrightarrow \underline{W}$ no longer changes over time

- $\underline{W}_i$ will / can differ from unit to unit

$$- \underline{W}_i^T \beta = \eta_i \text{ s.t. } Y_{it}(\infty) = \theta_t + \eta_i + e_{it}$$

s.t. θ_t is time fixed-effect

η_i is unit fixed-effect

Using panel for ATT:

$$Y_{iz}(\infty) = \theta_z + \eta_i + e_{iz}$$

$$Y_{i1}(\infty) = \theta_1 + \eta_i + e_{i1}$$

$$\Delta Y_{iz} = \Delta \theta_z + 0 + \Delta e_{iz}$$

$$E[\Delta e_z | G] = E[E[e_z | \underline{w}, G] | G] - E[E[e_1 | \underline{w}, G] | G]$$

$= 0$

$$\Rightarrow E[\Delta Y_z(\infty) | G=g] = \Delta \theta_z = E[\Delta Y_z(\infty)]$$

↳ parallel trends assumpt $\Rightarrow$ changes in level are same for treat, untreat

Group-time ATE:

$$ATT(g, t) = E[Y_t(g) - Y_t(\infty) | G=g]$$

$$ATT(g=2, t=2) = E[Y_2(2) - Y_2(\infty) | G=2]$$

$$= E[Y_2(2) - Y_1(\infty) | G=2] - E[Y_2(\infty) - Y_1(\infty) | G=2]$$

$$= E[\Delta Y_2 | G=2] - E[\Delta Y_2(\infty) | G=\infty]$$

$$= E[\Delta Y_2 | G=2] - E[\Delta Y_2 | G=\infty]$$

↳ Diff-in-Diff